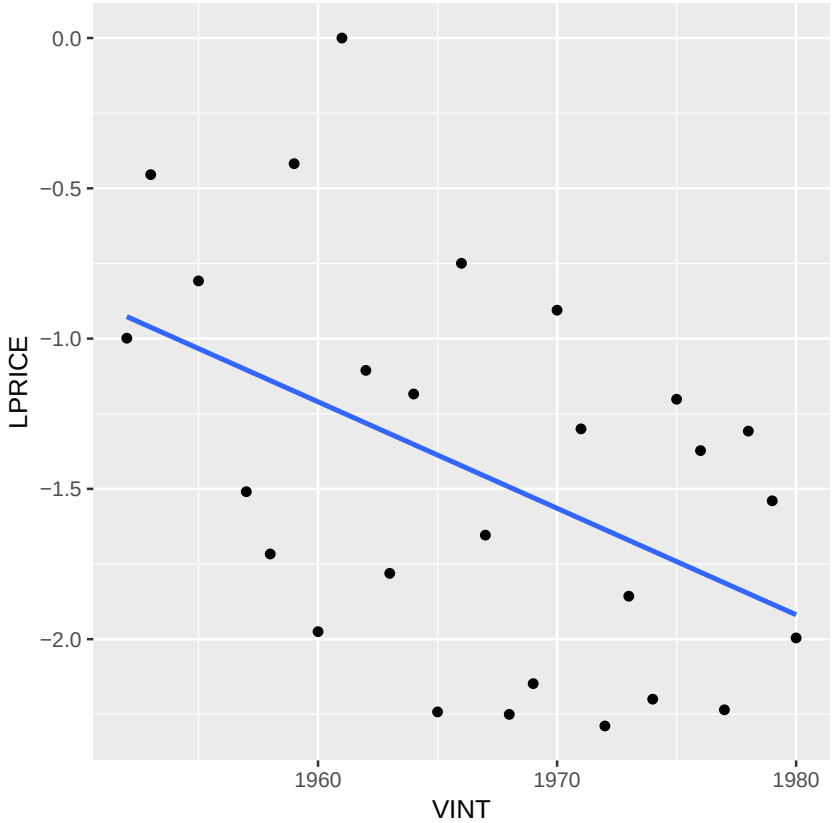
42.510 Statistical Learning for Data Science

Linear regression

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Week 4, Lecture 1

Linear regression



Problem setup

- n = Number of observations
- p = Number of predictor variables
- y = Dependent variable in $\mathbb{R}$
- $x_1, \ldots, x_p$ = Independent variables (predictors)

- Want to predict y with a function $f(\mathbf{x})$.
- Let $f(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p$.
- Data: $(y_i, x_{i1}, \ldots, x_{ip})$, $i = 1, \ldots, n$.
- Method of least squares:

$$\min_{\beta_0,\ldots,\beta_p} \frac{1}{n} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \ldots - \beta_p x_{ip})^2.$$

- This is Empirical Risk minimization (ERM).
- Minimize loss L where

$$R(f) = \mathbb{E}L(Y, f(\mathbf{X})) = \mathbb{E}(Y - f(\mathbf{X}))^2$$

with some assumptions on $\mathbb{P}(\mathbf{X}, Y)$.

- Structural assumption: $f(\mathbf{x}) = \mathbf{x}^\top \beta$.

Linear regression: Notation and set-up

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbb{X} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}.$$

- Optimal solution:

$$\widehat{\boldsymbol{\beta}} = (\mathbb{X}^\top \mathbb{X})^{-1} \mathbb{X}^\top \mathbf{y}.$$

- *Fitted values:* $\widehat{y}_i = \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}$.
- *Residuals:* $e_i = y_i - \widehat{y}_i$.
- Assume: $\mathbf{Y} = \mathbb{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$.
- Using t-test we answered: is X_i useful in predicting the response Y?
- We also discussed *Quality of fit* using R^2 and adjusted-R^2.

Linear regression: Quality of fit

- Recall:

$$\text{SSE} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2, \quad \text{SSR} = \sum_{i=1}^{n}(\hat{y}_i - \overline{y})^2, \quad \text{SST} = \sum_{i=1}^{n}(y_i - \overline{y})^2.$$

- We have the relationship:

$$\text{SST} \;=\; \text{SSR} \;+\; \text{SSE}$$

- Also recall *Coefficient of Determination* and *Adjusted-R^2*:

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}},$$

$$\text{Adjusted-}R^2 = 1 - (1 - R^2)\frac{n-1}{n-p-1}.$$

- An (unbiased) estimate for σ^2 is

$$\widehat{\sigma}^2 = \frac{1}{n-p-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{\text{SSE}}{n-p-1}.$$

Linear regression: other important questions

- Is the model significant?
  - Is at least one of the predictors $X_1, \ldots, X_p$ useful in predicting the response?
- Are all variables important?
  - Do all the predictors help to explain Y, or is only a subset of the predictors useful?
- How good is our prediction?
  - Given a set of predictor values, what response value should we predict, and how accurate is our prediction?

Linear regression: Quality of fit

Is the model significant?

- Digression: F-distribution and F-test for comparing variances and analysis of variance (ANOVA).

F-distribution and F-test

- **Set up:** Assume that $X_1, X_2, \ldots, X_{n_1}$ are iid from a $\mathcal{N}(\mu_1, \sigma_1^2)$ distribution, and $Y_1, Y_2, \ldots, Y_{n_2}$ are iid from a $\mathcal{N}(\mu_2, \sigma_2^2)$ distribution.
- Let

$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \overline{X}_{n_1})^2, \quad S_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (Y_i - \overline{Y}_{n_2})^2.$$

  In terms of realizations we have s_1^2 and s_2^2 respectively as sample variances.
- To test if the variances are the same, we try to see how far σ_1^2/σ_2^2 is away from 1.
- The test statistic will be in terms of s_1^2/s_2^2 which estimates σ_1^2/σ_2^2 .

- **Terminology:** when not all the random variables in a collection have the same variance, they are called **heteroscedastic**.

Snedecor's F distribution

- χ^2-**distribution:** If $Z_1, \ldots, Z_k$ are i.i.d. $\mathcal{N}(0,1)$, then $V = Z_1^2 + \ldots + Z_k^2$ follows a χ^2-distribution with k degrees of freedom (d.f.).
- Let $X \sim \chi^2_u$ and $Y \sim \chi^2_v$, X and Y are independent.
- Define the random variable

$$\mathrm{F} = \frac{X/u}{Y/v}.$$

- Then the random variable F follows an F-distribution with degrees of freedom u and v. We write $\mathrm{F} \sim F_{u,v}$.
- For a test (e.g., ANOVA) if the test statistic $\mathrm{F} \sim F_{u,v}$ under the null hypothesis $\mathbf{H}_0$, then we reject $\mathbf{H}_0$ if the observed test statistic value f_{obs} is very large or p-value is very small, where

$$\text{p-value} = \mathbb{P}(\mathrm{F} > f_{\text{obs}}).$$

The pdfs of the χ^2 and F distributions are omitted here, but you can find it in any Statistics textbook at the appropriate level.

Linear regression: Quality of fit

Is the model significant? (back from digression)

- We test goodness of fit of the linear regression model using an F-test.
- Here we test:

$$\mathbf{H}_0 : \beta_1 = \ldots = \beta_p = 0 \quad \text{vs.} \qquad \mathbf{H}_1 : \text{at least one } \beta_j \text{ is non-zero.}$$

- The test statistics is given by

$$\mathrm{F} = \frac{\mathrm{SSR}/p}{\mathrm{SSE}/(n-p-1)}.$$

- Under $\mathbf{H}_0$, we have $\mathrm{F} \sim F_{p,n-p-1}$.
- If the observed value of F is f_{obs}, then we reject $\mathbf{H}_0$ for large values of f_{obs}.
- At level α reject $\mathbf{H}_0$ if $f_{\text{obs}} > f_{\alpha,p,n-p-1}$ where $\mathbb{P}(X > f_{\alpha,p,n-p-1}) = \alpha$ for $X \sim F_{p,n-p-1}$. This is a one-sided test.

Linear regression: Quality of fit

- We can also test if only a particular subset of q coefficients are all zero.

$$\mathbf{H}_0 : \beta_{p-q+1} = \ldots = \beta_p = 0.$$

- Fit a second model using all variables except the last q. Suppose the sum of squares due to regression for the new model is SSR_{p-q}. Then the appropriate F-statistic is

$$\text{F}^* = \frac{(\text{SSR} - \text{SSR}_{p-q})/q}{\text{SSE}/(n-p-1)} = \frac{(\text{SSE}_{p-q} - \text{SSE})/q}{\text{SSE}/(n-p-1)},$$

  which under $\mathbf{H}_0$ follows an F-distribution with degrees of freedom $q, n-p-1$.

- If the observed value of F^* is f^*_{obs}, then we reject $\mathbf{H}_0$ for large values of f^*_{obs}.
- At level α reject $\mathbf{H}_0$ if $f^*_{\text{obs}} > f_{\alpha,q,n-p-1}$ where $\mathbb{P}(X > f_{\alpha,q,n-p-1}) = \alpha$ for $X \sim F_{q,n-p-1}$.

Linear regression: Are all variables important?

Model selection

- Least-squares is nice and elegant, but there are issues.
- *Prediction accuracy*: Low bias but Large variance.
  - **solution**: set some coefficients to zero.
  - we sacrifice a little bit of bias to reduce the variance.
- *Interpretation*: A smaller subset of predictors that exhibit the strongest effects helps to get the "big picture."

Subset selection

- Best subset selection, Forward and Backward step-wise selection. Often chosen with criterion *Mallow's* C_p, *AIC, BIC*, etc.
- Shrinkage methods.

We address these later in further details.

Predictions in regression models

- There are two kind of **predictions** in a *regression problem*.
  - Given $\mathbf{X} = \mathbf{x}_0$, estimate the **mean response**, i.e.,

$$\mathbb{E}[Y|\mathbf{X} = \mathbf{x}_0] = \mathbf{x}_0^\top \beta.$$

  - Predict the response for a particular observation where $\mathbf{X} = \mathbf{x}_0$, i.e., predict

$$Y = \mathbf{x}_0^\top \beta + \epsilon.$$

- In both cases the *point* estimate/prediction of the response is the same

$$\widehat{\mathbb{E}[Y|\mathbf{X} = \mathbf{x}_0]} = \widehat{Y} = \mathbf{x}_0^\top \widehat{\beta}.$$

- But the interval estimates are different.

Predictions in regression models

- For estimating the mean $\mathbb{E}[Y|\mathbf{X} = \mathbf{x}_0] = \mathbf{x}_0^\top \beta$ with $\mathbf{x}_0^\top \widehat{\beta}$, the variance is
$$\text{Var}(\mathbf{x}_0^\top \widehat{\beta}) = \mathbf{x}_0^\top (\mathbb{X}^\top \mathbb{X})^{-1} \mathbf{x}_0 \sigma^2.$$

- Since σ^2 is estimated by $\widehat{\sigma}^2 = \frac{1}{n-p-1} \sum_{i=1}^{n} (y_1 - \hat{y}_i)^2$, a $100(1-\alpha)\%$ **confidence interval** is:
$$\mathbf{x}_0^\top \widehat{\beta} \pm t_{n-p-1,\alpha/2} \widehat{\sigma} \sqrt{\mathbf{x}_0^\top (\mathbb{X}^\top \mathbb{X})^{-1} \mathbf{x}_0}.$$

- For estimating $Y = \beta^\top \mathbf{x}_0 + \epsilon$ with $\mathbf{x}_0^\top \widehat{\beta}$, the variance is
$$\text{Var}(\mathbf{x}_0^\top \widehat{\beta} + \epsilon) = \left(1 + \mathbf{x}_0^\top (\mathbb{X}^\top \mathbb{X})^{-1} \mathbf{x}_0\right) \sigma^2.$$

- Hence a $100(1-\alpha)\%$ **prediction interval** is:
$$\mathbf{x}_0^\top \widehat{\beta} \pm t_{n-p-1,\alpha/2} \widehat{\sigma} \sqrt{1 + \mathbf{x}_0^\top (\mathbb{X}^\top \mathbb{X})^{-1} \mathbf{x}_0}.$$

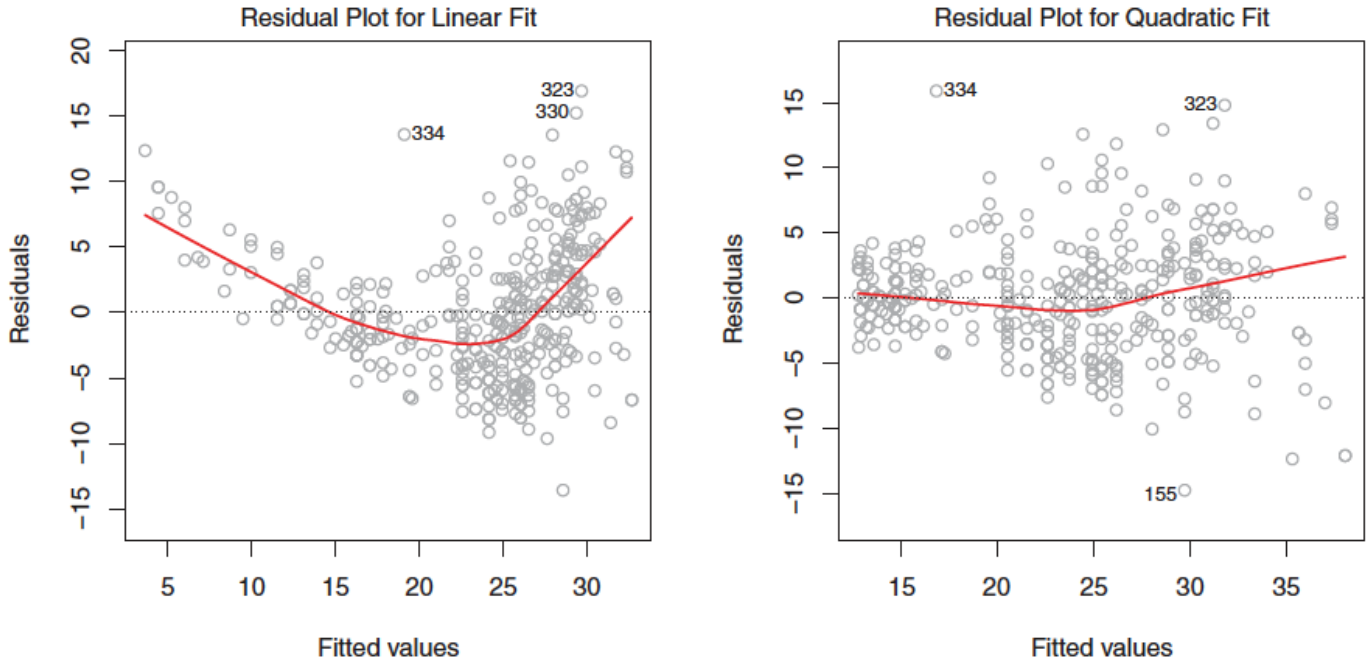
- A prediction interval is always wider than its corresponding confidence interval.

Potential concerns in linear regression

- Non-linearity
- Correlation in error terms
- Outliers
- Heteroscedasticity
- Collinearity

Non-linearity

- Linear regression assumes **linearity** between y and $\mathbf{x}$.
- **Residual plot**: Plot residuals $e_i = y_i - \hat{y}_i$ vs. fitted/predicted values $\hat{y}_i$.



- Left: regress `mpg` on `horsepower`.
- Right: regress `mpg` on `horsepower`+ `horsepower`^{2}.
- Try non-linear transformation of predictors: $\sqrt{X}, X^2, \log X$.

Correlation in error terms

- Linear regression assumes i.i.d. errors, hence ϵ_i's are **uncorrelated**.
- **Issue**: Underestimation of both confidence and prediction intervals.
- A 95% CI or PI may in reality have smaller confidence.
- Residual plot.
- In time-series data there exist methods to account for this.
- Good experimental design also helps in mitigating risk of correlated error terms.

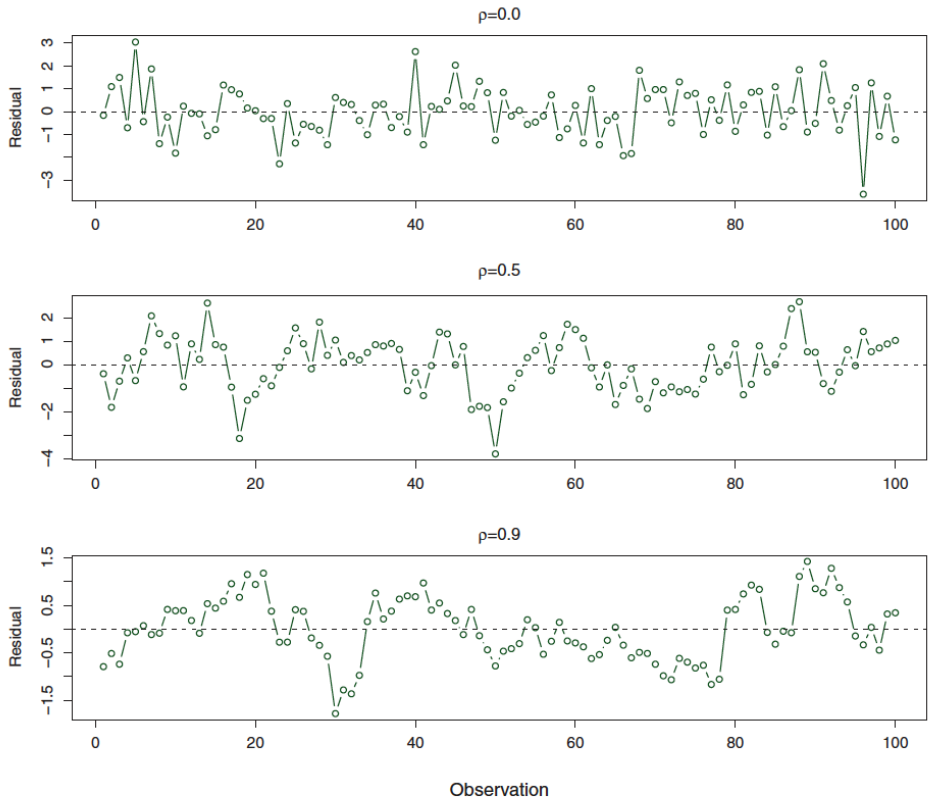
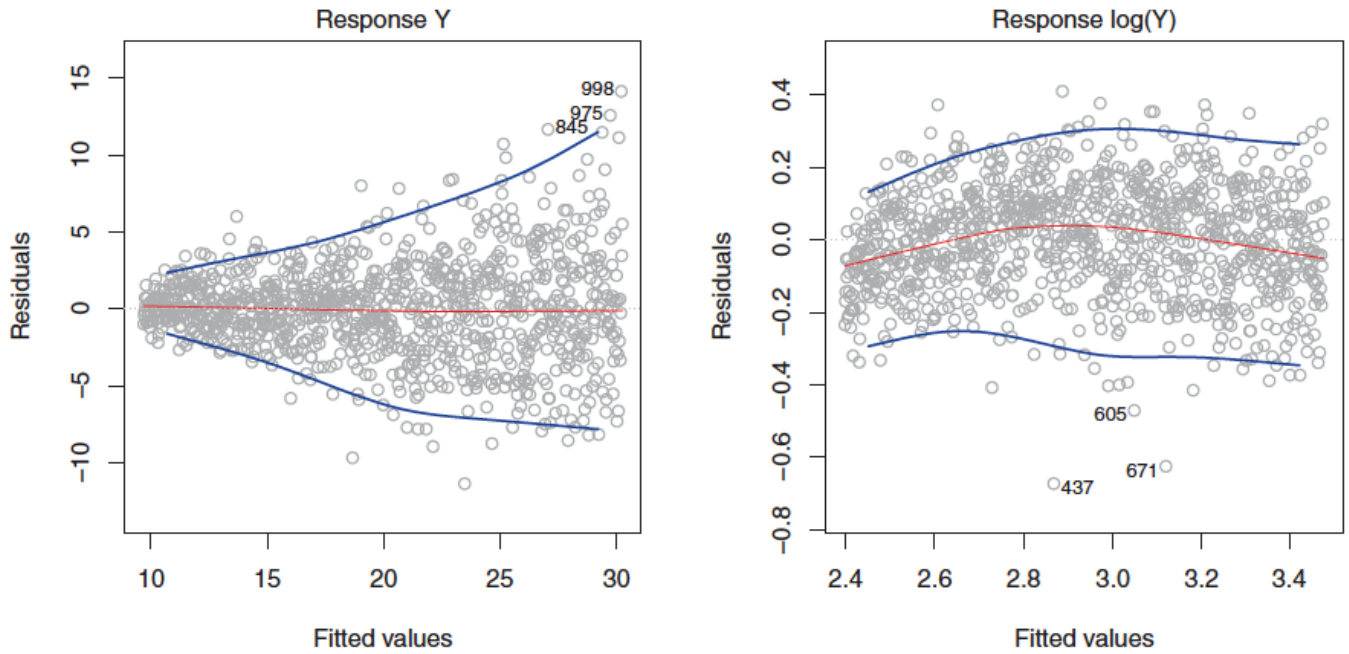


FIGURE 3.10. *Plots of residuals from simulated time series data sets generated with differing levels of correlation ρ between error terms for adjacent time points.*

Heteroscedasticity (non-constant error)

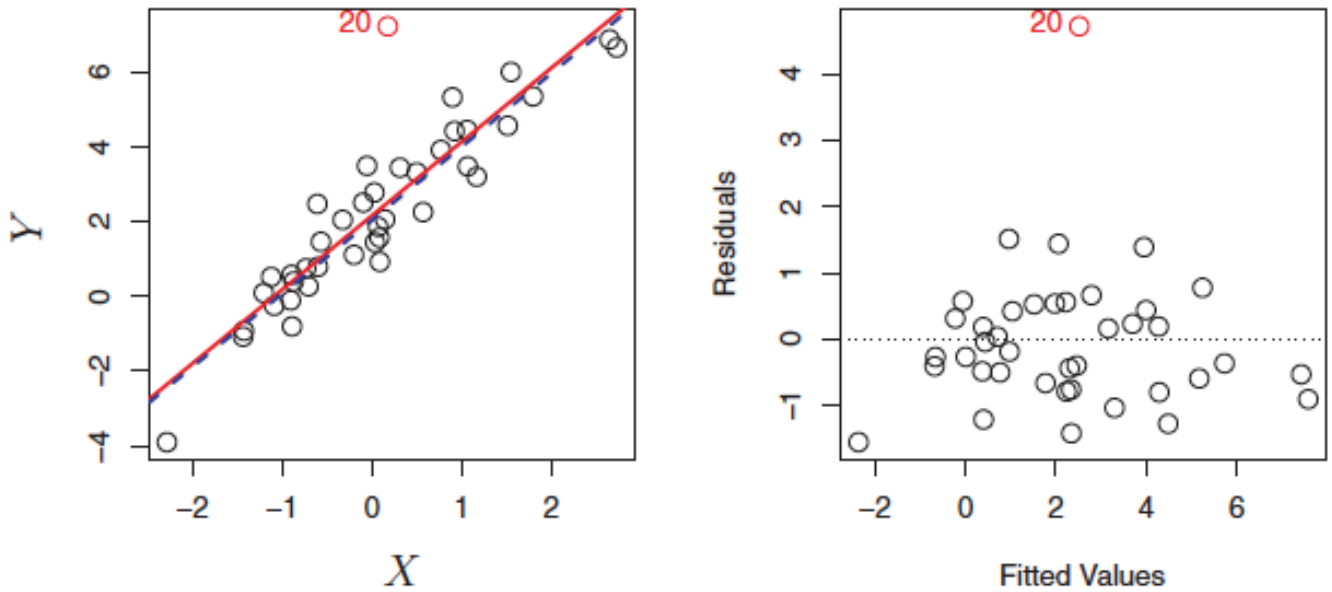
- Linear regression assumes i.i.d. errors, hence ϵ_i's have **constant variance**.
- Residual plot.



- Possible solution:
  - transform the response Y with a concave function $\log Y, \sqrt{Y}$.
  - Fit weighted least squares with weights proportional to inverse of variance (if variance is known, or its structure understood).

Outliers

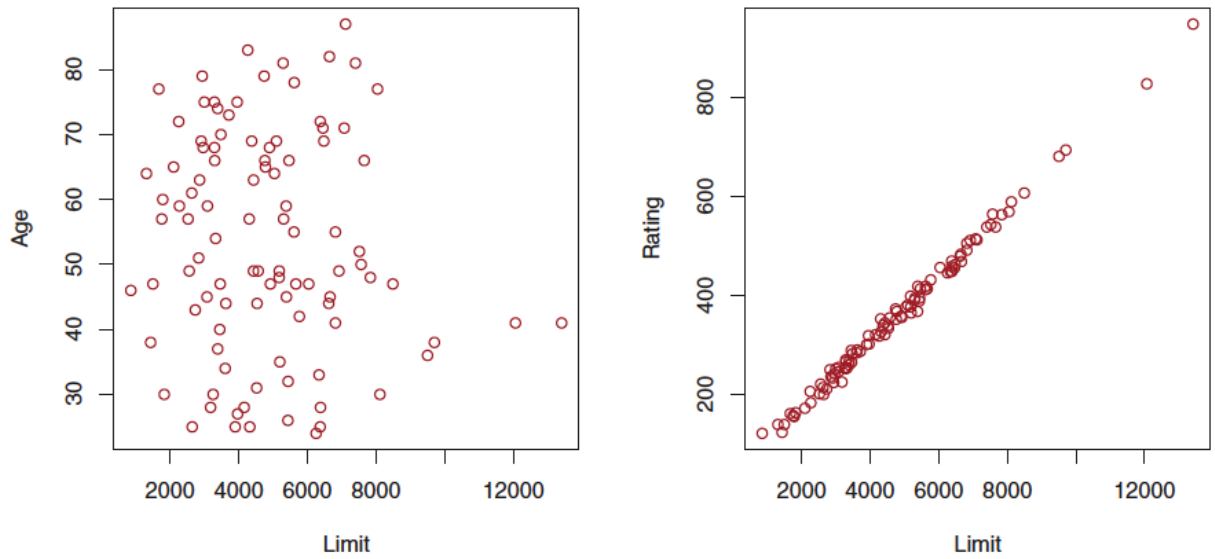
- **Outlier** is a response value far from its model predicted value.
- Residual plot



- Blue regression line after removing residual; not much difference.
- Still there may be difference in RMSE and R^2.
- Studentization of residuals to check for outliers.
  - Divide residual e_i by its estimated standard error.
  - Check if *studentized residual* values are between -3 and 3.

Collinearity

- **Collinearity** or **multicollinearity**: two or more predictor variables are closely related to one another (possibly linearly).
- Correlation matrix and/or **Scatter plot**.



- For multicollinearity compute *variance inflation factor*. (omitted)
- Solution:
  - drop problematic variables.
  - combine collinear variables to a single predictor variable.

References

- Hastie et al. (2001), *The Elements of Statistical Learning*, Chapter 3.
- James et al. (2023), *Introduction to Statistical Learning*, Chapter 3.

F-distribution and F-test: comparing variances (optional)

- Note that $\frac{(n_i-1)S_i^2}{\sigma_i^2}$ is a $\chi^2_{n_i-1}$ random variable for $i = 1, 2$.
- Hence $\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1,\, n_2-1}$.
- Hence

$$\Pr\left(f_{1-\alpha/2,\, n_1-1,\, n_2-1,} \leq \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \leq f_{\alpha/2,\, n_1-1,\, n_2-1,}\right) = 1-\alpha,$$

  where $\Pr(\mathrm{F} \geq f_{\beta, n_1-1,\, n_2-1}) = \beta$ where $\mathrm{F} \sim F_{n_1-1,\, n_2-1}$.
- In R use: $f_{\beta, n_1-1,\, n_2-1}$ =qf$(1-\beta, n_1-1, n_2-1)$.
- Therefore, a two-sided $100(1-\alpha)\%$ **confidence interval** (CI) for σ_1^2/σ_2^2 is

$$\left[\frac{1}{f_{\alpha/2,\, n_1-1,\, n_2-1,}}\frac{s_1^2}{s_2^2},\ \frac{1}{f_{1-\alpha/2,\, n_2-1,\, n_1-1}}\frac{s_1^2}{s_2^2}\right].$$

Here s_1, s_2 are the observed versions of the random variables S_1, S_2.

Exercise: comparing variances (optional)

- The test statistic for the hypothesis test is $s = s_1^2/s_2^2$ with the two-tailed rejection region:

$$f \leq f_{1-\alpha/2,\, n_1-1,\, n_2-1} \quad \text{or} \quad f \geq f_{\alpha/2,\, n_1-1,\, n_2-1}$$

- Suppose we observe $s_1 = 1.1$, $s_2 = 0.9$, and $n_1 = n_2 = 50$.
- Test $\mathbf{H}_0 : \sigma_1^2 = \sigma_2^2$ vs $\mathbf{H}_A : \sigma_1^2 \neq \sigma_2^2$, using $\alpha = 0.05$?
- We have using R:
  - $f_{0.025,49,49} = \texttt{qf(0.975,49,49)} = 1.762$,
  - $f_{0.975,49,49} = \texttt{qf(0.025,49,49)} = 0.567$.
- Hence the 95% CI for σ_1^2/σ_2^2 is [0.848, 2.63].
- We cannot reject $\mathbf{H}_0$ at 5% level of significance since 1 is in the CI.